



EXAMINING THE EFFECTIVENESS OF DIFFERENT COATINGS AT PREVENTING CORROSION OF A GALVANISED STEEL SCREW WHEN EXPOSED TO ACID RAIN

Subject: Chemistry
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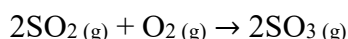
Research Question

How effective are different coatings (chain lubricant, vegetable oil, coconut oil, WD-40 and no coating) at preventing corrosion of a galvanised steel screw when exposed to 0.1 mol dm^{-3} sulfuric acid, as measured by the percentage difference in the screw's width, length and mass using a digital calliper and an electronic balance?

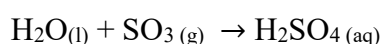
Introduction

Corrosion of a metal is the deterioration of its physical properties caused by a reaction with its environment, allowing the metal to convert into a more stable compound, such as a metal sulfide or a metal oxide (Montgomery, 2022). Corrosion has major global impacts, costing nearly \$US 2.9 trillion (NDE Solutions, 2022), due to the weakening of metals in infrastructure, such as bridges and automobiles. By studying the effectiveness of various protective coatings, this investigation aims to find more efficient, low-cost and more readily available solutions to reduce corrosion and extend the lifespan of metal products. As sustainability is prioritised, understanding which solutions can protect metals from corroding will reduce the need for frequent replacements of rusted metals, decreasing resource consumption and waste. Furthermore, ports produce sulfur oxides (Mueller et al., 2023), and go-kart drivers who race near a port deal with acid rain; hence, knowing which coatings can protect go-karts from corrosion is of great significance.

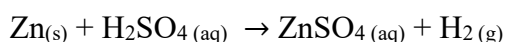
Galvanised steel screws have been selected as they are the most common screw found on go-karts. These iron screws have a protective zinc coating to prevent corrosion, but prolonged exposure to acid rain wears down the coating, resulting in corrosion of the iron. To simulate acid rain for this experiment, 0.1 mol dm^{-3} sulfuric acid will be used as the sulfur dioxide released from the ports will first react with the oxygen in the air to form sulfur trioxide.



Once the sulfur trioxide has been formed, it will react and dissolve in rainwater to form sulfuric (VI) acid (Brown et al., 2023/2023, p. 312).

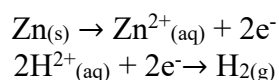


As these are galvanised steel screws, the sulfuric acid will first react with the zinc coating which produces flammable hydrogen gas.

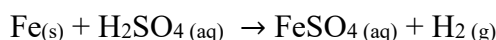


The above equation shows that when zinc reacts with sulfuric acid, the hydrogen atoms are displaced and form zinc sulfate. The displacement of the hydrogen atoms refers to the hydrogen atom being replaced by an active metal, in this case, zinc. An active metal is a metal that readily undergoes chemical reaction (Editorial team of the PCC Chemical Academy, 2024). Zinc sulfate is a colourless compound that is formed through zinc's oxidation in the presence of sulfate ions.

This reaction is a redox reaction, which is an electron transfer reaction, as the zinc is oxidised, and the hydrogen ions are reduced, which can be seen in the half equations below.



Once the zinc layer is depleted, the underlying iron is exposed and begins to corrode:



To prevent the zinc layer from depleting and the underlying iron from corroding, four coatings will be tested: chain lubricant, vegetable oil, coconut oil and WD-40.

0.1 mol dm⁻³ sulfuric acid was used to simulate the long-term corrosive effects of acid rain, where acid rain may become trapped on screws, resulting in continuous reactions. Since acid rain has a pH of 4, it corrodes metal slowly, hence 0.1 mol dm⁻³ sulfuric acid which has a pH of 0.98 (JRG Beavon, 2016) was used to accelerate the corrosion process, allowing observable effects after 14 days. This method of accelerated corrosion testing maintains chemical relevance while making long-term effects measurable in a short timeframe.

WD-40 is a petrol-based solution that contains hydrocarbons (Kowalski, n.d.), making it effective at displacing water and creates a thin hydrophobic layer over the screw. A hydrophobic layer repels water (Macios, 2024), not allowing the water to corrode the iron screw. As hydrocarbons are non-polar molecules, in contrast to sulfuric acid, which is polar, it would allow WD-40 to be effective in reducing corrosion as the non-polar hydrocarbons would repel the sulfuric acid, which limits the corrosion of the screw. A key component of WD-40 is petrol, it relies on the use of crude oil to manufacture, which is a sustainability issue. Extraction of crude oil releases greenhouse gases such as methane, leading to environmental damage.

Chain lubricant is a go-kart-chain-specific chemical that is designed to repel water and resist oxidation, similar to WD-40. Chain lubricant is primarily made of hydrocarbons such as propane, butane, and naphtha petroleum. This is different to WD-40 as naphtha petroleum usually has carbon chains up to 12 carbons long, unlike the crude oil in WD-40, which has carbon chains up to more than 40 carbons long. This difference means that chain lubricant has weaker intermolecular forces, making it evaporate easily in comparison to WD-40 (Takahashi et al., 2022).

Vegetable oil is an oil commonly found in households and is a natural, biodegradable coating that serves to limit wastage and promote sustainability. Vegetable oil contains triglycerides, which are esters in which three molecules of the same or different fatty acids are bonded to an alcohol glycerol. While alcohols and esters are polar, the dominant long carbon chain in triglycerides makes the molecule non-polar (Soult, 2016). As vegetable oil is non-polar like WD-40, this investigation will produce data that will illustrate whether household items are more effective at preventing corrosion in comparison to items designed to prevent corrosion.

Coconut oil has the same composition as vegetable oil, however, the key difference between coconut oil and vegetable oil is that coconut oil is more viscous (Thompson, 2024), and it should create a more protective coat over the screw to repel the sulfuric acid from reacting and corroding the screw.

A control group using uncoated galvanized steel screws was used to provide a baseline for extent of corrosion in the absence of any protective layer.

Variables

Independent Variable

The independent variable is the coating that will be applied on the screw being: WD-40, chain lubricant, vegetable oil, coconut oil, and no coating.

Dependent Variable

The amount of corrosion will be measured in three aspects: the corrosion of the length of the screw, measured by digital callipers ($\pm 0.01\text{mm}$), the corrosion of the width of the screw head, measured by digital callipers, and the corrosion of the mass of the screw, measured by an electronic balance ($\pm 0.01\text{g}$).

Controlled Variables

Variable	Reason for controlling	Method to control
Amount of coating	All screws are equally coated in the same amount of solution.	Measure 50 cm^3 of each coating for each screw.
Time the screws are soaked in the coating	All screws receive uniform adhesion of coating.	Ensured all screws are soaked in the coating for the same amount of time (30 minutes). This will allow each coating sufficient time to adhere uniformly to the surface of the screw while avoiding excessive buildup.
Time the screws are soaked in sulfuric acid	All screws face the same conditions to understand what prevents corrosion.	Ensured all screws are soaked in sulfuric acid for the same amount of time (14 days). Since a very low concentration of sulfuric acid is used, leaving the screws to soak for fewer days may not cause visible or measurable corrosion. Allowing the screws to soak for more time may lead to extensive corrosion especially to the control, possibly making it more difficult to take measurements.
Temperature	All screws face the same conditions and so that some reactions do not face a faster rate of reaction than others.	Ensure all screws were kept in the same area and faced fairly similar conditions with near constant temperature.
Volume and concentration of sulfuric acid that screws were submerged in	All screws face the same conditions to understand what prevents corrosion.	Each screw was submerged in 50 cm^3 of 0.1 mol dm^{-3} sulfuric acid.

Table 1. Controlled variables for the investigation, including the reason and the method by which it was controlled.

Materials

Glassware

- 1x 500 cm³ conical flask
- 25x 100 cm³ beakers
- 5x 50 cm³ graduated cylinders (± 0.75 cm³)
- Funnel

Chemicals

- 40 cm³ WD-40
- 40 cm³ chain lubricant
- 40 cm³ vegetable oil
- 40 cm³ melted coconut oil
- 500 cm³ deionised water
- 1250 cm³ 0.1 mol dm⁻³ sulfuric acid (H₂SO₄)

Equipment

- 1x stopwatch
- 5x Petri dishes (with lids)
- 25x galvanised steel screws
- 1x spatula
- 1x electronic balance (± 0.01 g)
- 1x calliper (± 0.01 mm)
- 5x papers
- 10x paper towels
- 1x tweezers
- 1x permanent marker
- 1x fume cupboard

Method

Part I

1. Take the measurements of each screw, including each screw's mass using an electronic balance and callipers to measure the length of the screw and the width of the screw head as depicted in Figure 1.

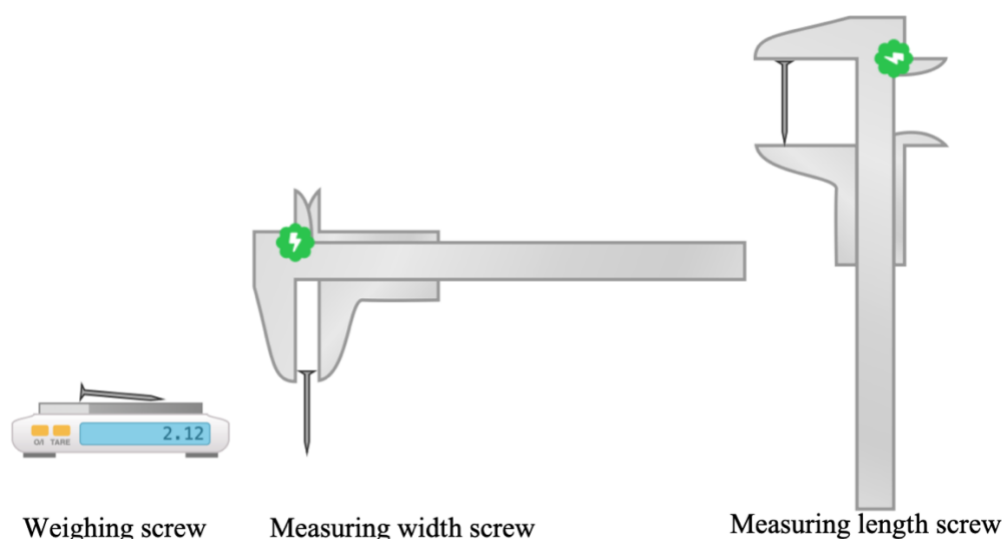


Figure 1. Obtaining measurements of the screw (Image made using chemix.org)

2. Label 5 Petri dishes with *vegetable oil*, *coconut oil*, *WD-40*, *chain lubricant*, and *control*.

3. Prepare the *vegetable oil* Petri dish by measuring 40 cm³ of vegetable oil through the use of a graduated cylinder and pouring it into the Petri dish.
4. Follow the same procedure for the other coatings
5. Prepare the *control* Petri dish by ensuring it is empty.
6. Once all Petri dishes are prepared, using tongs, place 5 screws into each Petri dish and ensure all screws are not touching and are submerged as illustrated in Figure 2.



Top view of the Petri dish



Side view of the Petri dish

Figure 2. Orientation of screws in Petri dish when submerged in coating (Image made using chemix.org)

7. Leave the screws submerged in the contents of the Petri dish for 30 minutes.
8. After 30 minutes, using tweezers, remove the set of screws from the beaker and place on a paper towel. Do not rub off the coating. Gently pat down the screws ensuring that any excess coating is removed.

Part II

1. Label 25 100 cm³ beakers with numbers 1 to 25.
2. Measure 50 cm³ of 0.1 mol dm⁻³ sulfuric acid using a graduated cylinder and pour 50 cm³ into each beaker.
3. Place all the beakers into a fume cupboard.
4. Leave the screws submerged in sulfuric acid for 14 days.

Part III

1. After 14 days, carefully remove the screws using tweezers.
2. Rinse using deionised water.
3. Use paper towels to dry the screws.
4. Measure the mass of each screw using a precision scale.
5. Using callipers, measure the length of the screw and the width of the screw head as depicted in figure 1.

Safety, ethical and environmental considerations

Safety considerations

- Wear safety goggles, a lab coat and gloves to protect eyes and skin from sulfuric acid and possible irritation from WD-40 and/or the chain lubricant.
- Ensure all glassware is crack-free and is not defective to prevent spilling, leaking or cuts.
- Hydrogen gas, a flammable product of the reaction between sulfuric acid and zinc, is released during the experiment. To reduce the risk of ignition, all beakers were placed in a fume cupboard so that the hydrogen gas would not be exposed to any heat sources.

Ethical considerations

- There are no relevant ethical considerations with this investigation.

Environmental considerations

- 0.1 mol dm^{-3} of sulfuric acid was poured down the drain but was neutralised by keeping the tap flowing as the sulfuric acid was drained, diluting the concentration of the sulfuric acid.
- Since WD-40 and chain lubricants are petrol-based solutions, they were disposed of in a chemical waste bin.
- Coconut oil and vegetable oil were not poured down the drain. Instead, they were discarded with regular garbage once the oils had solidified. This prevents the oils from solidifying in pipes, resulting in blockages.

Results

Qualitative Data

There were bubbles around each screw as soon as they were placed in the sulfuric acid. Layer of bubbles at the top of the beakers which had screws coated in vegetable oil and coconut oil. After the 14 days, all the beakers had an orange solution with zinc sulfate being produced, but all the threads of the screws seemed unaffected.

Raw Data

Coating	Trial 1: Length of screw corroded $l_c (coating)$ $\pm 0.01\text{mm}$	Trial 2: Length of screw corroded $l_c (coating)$ $\pm 0.01\text{mm}$	Trial 3: Length of screw corroded $l_c (coating)$ $\pm 0.01\text{mm}$	Trial 4: Length of screw corroded $l_c (coating)$ $\pm 0.01\text{mm}$	Trial 5: Length of screw corroded $l_c (coating)$ $\pm 0.01\text{mm}$
Vegetable Oil	0.24	0.29	0.32	0.27	0.30
WD-40	0.32	0.32	0.32	0.33	0.36
Chain Lubricant	0.23	0.23	0.23	0.27	0.27
Coconut Oil	0.23	0.20	0.21	0.24	0.24
No Coating	0.31	0.36	0.32	0.30	0.37

Table 2. Length of the screw that has been corroded for each coating type, with all having the same uncertainty of $\pm 0.01\text{m}$ as it is the uncertainty of the digital calliper.

Coating	Trial 1: Width of screw corroded $W_c (coating)$ $\pm 0.01\text{mm}$	Trial 2: Width of screw corroded $W_c (coating)$ $\pm 0.01\text{mm}$	Trial 3: Width of screw corroded $W_c (coating)$ $\pm 0.01\text{mm}$	Trial 4: Width of screw corroded $W_c (coating)$ $\pm 0.01\text{mm}$	Trial 5: Width of screw corroded $W_c (coating)$ $\pm 0.01\text{mm}$
Vegetable Oil	0.13	0.12	0.10	0.12	0.09
WD-40	0.05	0.05	0.05	0.05	0.05
Chain Lubricant	0.11	0.12	0.09	0.11	0.12
Coconut Oil	0.13	0.12	0.14	0.16	0.13
No Coating	0.18	0.17	0.20	0.20	0.18

Table 3. Width of the screw that has been corroded for each coating type, with all having the same uncertainty of $\pm 0.01\text{m}$ as it is the uncertainty of the digital calliper.

Coating	Trial 1: Mass of screw corroded $m_c(\text{coating})$ $\pm 0.005\text{g}$	Trial 2: Mass of screw corroded $m_c(\text{coating})$ $\pm 0.005\text{g}$	Trial 3: Mass of screw corroded $m_c(\text{coating})$ $\pm 0.005\text{g}$	Trial 4: Mass of screw corroded $m_c(\text{coating})$ $\pm 0.005\text{g}$	Trial 5: Mass of screw corroded $m_c(\text{coating})$ $\pm 0.005\text{g}$
Vegetable Oil	0.25	0.25	0.25	0.25	0.25
WD-40	0.26	0.25	0.26	0.26	0.26
Chain Lubricant	0.26	0.26	0.26	0.25	0.26
Coconut Oil	0.24	0.24	0.24	0.25	0.24
No Coating	0.24	0.28	0.26	0.26	0.26

Table 4. Mass of the screw that has been corroded for each coating type, with all having the same uncertainty of $\pm 0.005\text{g}$ as it is the uncertainty of the electron balance.

Data Processing

Using an absolute value for the change in mass and length does not provide the complete picture of which coating is best at preventing corrosion as each screw has different starting dimensions. Variations in initial mass would result in a larger screw naturally losing more material than a smaller one, even if the proportional corrosion is the same. As such, comparing raw changes without considering size and mass differences can result in misleading conclusions. A better approach would be to calculate the percentage change in mass, length and width of the screw, which will normalise the data relative of the original dimensions of each screw. This will allow for a more accurate comparison across all screws, and which coating was most effective at limiting corrosion. Below is a sample calculation that will be done to obtain the percentage difference, including the uncertainty associated with the data value. Screw 1 will be used as an example for the calculations. It was the screw with vegetable oil as the coating, refer to the appendix for the data table from which the data value was taken (highlighted in green).

$$\%_{\text{difference}} = \frac{\text{initial length} - \text{final length}}{\text{initial length}} \times 100 \pm 0\%$$

$$\%_{\text{difference}} = \frac{37.38 \pm 0.01 - 37.14 \pm 0.01}{37.38 \pm 0.01} \times 100 \pm 0\%$$

$$\%_{\text{difference}} = \frac{0.24 \pm 0.02}{37.38 \pm 0.01} \times 100 \pm 0$$

To continue, all the absolute uncertainties must be converted to percentage uncertainties. Below is a sample calculation for the uncertainty of the difference in length; the same process will be applied to the initial length.

$$0.24 \pm 0.02 = 0.24 \pm \left(\frac{0.02}{0.24} \times 100 \pm 0 \right)$$

$$0.24 \pm 0.02 = 0.24 \pm 8.33\%$$

$$\%_{\text{difference}} = \frac{0.24 \pm 8.33\%}{37.38 \pm 0.03\%} \times 100 \pm 0\%$$

$$\%_{\text{difference}} = 0.0064 \pm 8.36\% \times 100 \pm 0\%$$

$$\%_{\text{difference}} = 0.64 \pm 8.36\%$$

As this is a percentage uncertainty in the percentage difference, it can be confusing to interpret. Therefore, the absolute uncertainty in the percentage difference has been calculated to provide a clearer representation. Below is a sample calculation.

$$\%_{\text{difference}} = (0.64 \pm 0.053)\%$$

$$\%_{\text{difference}} = (0.64 \pm 0.05)\%$$

The same calculations will be applied to all the data values, which can be seen in Tables 5, 6 and 7.

The interquartile range (*IQR*) method was used to determine outliers, as it is most effective for analysing smaller data sets compared to the standard deviation method. Below is the method, and using Excel, all of the data points were processed, resulting in outliers, which are highlighted in yellow in Tables 5, 6 and 7.

$$Q1 \text{ (First Quartile)} = \text{Lower half of median}$$

$$Q3 \text{ (Third Quartile)} = \text{Upper half of median}$$

$$IQR \text{ (Interquartile Range)} = Q3 - Q1$$

$$\text{Lower bound} = Q1 - (1.5 \times IQR)$$

$$\text{Upper bound} = Q3 + (1.5 \times IQR)$$

A data point that is greater than the upper bound or less than the lower bound is considered an outlier.

Coating	Trial 1: Percentage of the screw's length that has corroded $\%l_c (coating)$ $\pm 0.05\%$	Trial 2: Percentage of the screw's length that has corroded $\%l_c (coating)$ $\pm 0.05\%$	Trial 3: Percentage of the screw's length that has corroded $\%l_c (coating)$ $\pm 0.05\%$	Trial 4: Percentage of the screw's length that has corroded $\%l_c (coating)$ $\pm 0.05\%$	Trial 5: Percentage of the screw's length that has corroded $\%l_c (coating)$ $\pm 0.05\%$
Vegetable Oil	0.24	0.29	0.32	0.27	0.30
WD-40	0.32	0.32	0.32	0.33	0.36
Chain Lubricant	0.23	0.23	0.23	0.27	0.27
Coconut Oil	0.23	0.20	0.21	0.24	0.24
No Coating	0.31	0.36	0.32	0.30	0.37

Table 5. Percentage of the screw's length that has been corroded for each coating type, with all having the same uncertainty of $\pm 0.05\%$.

Coating	Trial 1: Percentage of the screw's width that has corroded $\%w_c (coating)$ %	Trial 2: Percentage of the screw's width that has corroded $\%w_c (coating)$ %	Trial 3: Percentage of the screw's width that has corroded $\%w_c (coating)$ %	Trial 4: Percentage of the screw's width that has corroded $\%w_c (coating)$ %	Trial 5: Percentage of the screw's width that has corroded $\%w_c (coating)$ %
Vegetable Oil	1.75 $\pm$ 0.27	1.65 $\pm$ 0.28	1.35 $\pm$ 0.27	1.63 $\pm$ 0.27	1.20 $\pm$ 0.27
WD-40	0.69 $\pm$ 0.28	0.67 $\pm$ 0.27	0.67 $\pm$ 0.27	0.68 $\pm$ 0.27	0.68 $\pm$ 0.27
Chain Lubricant	1.47 $\pm$ 0.27	1.62 $\pm$ 0.27	1.19 $\pm$ 0.27	1.49 $\pm$ 0.27	1.63 $\pm$ 0.27
Coconut Oil	1.76 $\pm$ 0.27	1.60 $\pm$ 0.27	1.87 $\pm$ 0.27	2.15 $\pm$ 0.27	1.72 $\pm$ 0.27
No Coating	2.38 $\pm$ 0.27	2.23 $\pm$ 0.26	2.67 $\pm$ 0.27	2.70 $\pm$ 0.27	2.30 $\pm$ 0.26

Table 6. Percentage of the screw's width that has been corroded for each coating type, with all having different uncertainties.

Coating	Trial 1: Percentage of the screw's mass that has corroded $\%m_c (coating)$ %	Trial 2: Percentage of the screw's mass that has corroded $\%m_c (coating)$ %	Trial 3: Percentage of the screw's mass that has corroded $\%m_c (coating)$ %	Trial 4: Percentage of the screw's mass that has corroded $\%m_c (coating)$ %	Trial 5: Percentage of the screw's mass that has corroded $\%m_c (coating)$ %
Vegetable Oil	11.36±0.48	11.42±0.48	11.21±0.47	11.26±0.48	11.31±0.48
WD-40	11.76±0.48	11.31±0.48	11.56±0.47	11.71±0.48	11.87±0.48
Chain Lubricant	11.71±0.48	11.87±0.48	11.87±0.48	11.31±0.48	11.76±0.48
Coconut Oil	11.16±0.49	10.91±0.48	10.96±0.48	11.21±0.47	10.91±0.48
No Coating	11.06±0.49	12.56±0.48	11.93±0.49	11.76±0.48	11.82±0.48

Table 7. Percentage of the screw's mass that has been corroded for each coating type, with all having different uncertainties.

Using the raw data collected in Table 6, a set of data can be developed for the average of the trials, as it will be able to get a more accurate value. A sample calculation is below, taken from the data of the coating of WD-40 (highlighted in Table 6). The other trials will be processed automatically in Excel using this method.

$$\%w_{c(WD-40)} = \frac{(0.69 \pm 0.28) + (0.67 \pm 0.27) + (0.67 \pm 0.27) + (0.68 \pm 0.27) + (0.68 \pm 0.27)}{(5 \pm 0)}$$

$$\%w_{c(WD-40)} = \frac{(3.39 \pm 1.36)}{(5 \pm 0)}$$

$$\%w_{c(WD-40)} = \frac{(3.39 \pm 40.118\%)}{(5 \pm 0\%)}$$

$$\%w_{c(WD-40)} = 0.678 \pm 40.118\%$$

$$\%w_{c(WD-40)} = (0.68 \pm 0.27)\%$$

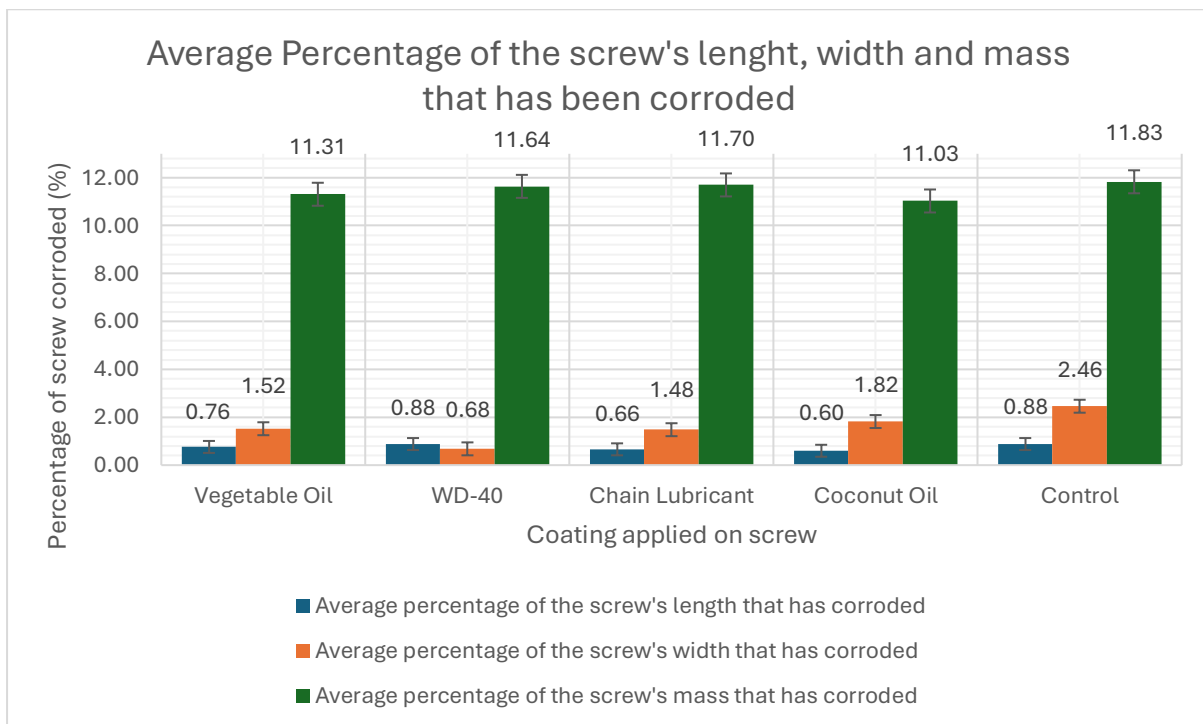
The same steps will be taken to calculate the percentage uncertainty for the width corroded and the mass corroded. Calculations for the rest of the results with their uncertainties using an Excel spreadsheet are shown in Table 8.

Processed Data

Coating	Average percentage of the screw's length that has corroded $\pm 0.25\%$	Average percentage of the screw's width that has corroded $\pm 0.27\%$	Average percentage of the screw's mass that has corroded $\pm 0.48\%$
Vegetable Oil	0.76	1.52	11.31
WD-40	0.88	0.68	11.64
Chain Lubricant	0.66	1.48	11.70
Coconut Oil	0.60	1.82	11.03
No Coating	0.88	2.46	11.83

Table 8. Average percentage of the screw's length, width and mass that has been corroded for each coating type, with all having constant uncertainties for each aspect of corrosion.

The data from Table 8 will be plotted onto a bar graph to get a visual representation of the data to understand which coating is the most effective.



Graph 1. Graphical representation of the average percentage of the screw's length, width and mass that has been corroded alongside which coating was applied on the screw.

Conclusion

The presence of outliers, despite not being removed, may have impacted the average percentage of corrosion, potentially influencing which coating was most effective. However, as there were limited outliers, their impact on the data is likely minimal.

The results suggest that all four coatings provided some protection against corrosion when compared to the control, which experienced the highest average corrosion across all three measured aspects. Coconut oil showed to be the most effective overall protective coating, with the lowest average mass loss of 11.03% and the lowest average corrosion of the screw's length of 0.60%. Chemically, coconut oil contains triglycerides, which form a viscous and non-polar barrier. The increased viscosity likely allowed for a thicker, more uniform coating, which effectively repelled the polar sulfuric acid, reducing exposure to oxygen. Since vegetable oil is less viscous, it likely formed a thinner barrier than coconut oil, resulting in outperforming the control across all dimensions, but less effectively than coconut oil.

As the error bars from Graph 1 were similar across all coatings, they were unlikely to have significantly affected the comparative effectiveness of the results.

Chain lubricant had the least average corrosion of the screw's width of 0.68%, nearly four times lower than the control. The short-chain hydrocarbons, like propane in chain lubricant, created a volatile but effective hydrophobic layer. However, due to its high volatility and low viscosity, it may have caused it to evaporate quickly, explaining the higher mass loss percentage.

In contrast, WD-40, though designed for corrosion prevention, showed no reduction in the corrosion of screw's length as it had the same percentage of length corroded as the control of 0.88%. This could suggest that the WD-40 was not applied evenly across the length of the screw. However, in the future, a more standardised method, such as dipping or controlled spraying, can be used to ensure even application of each coating, reducing the uncertainties. It is possible that WD-40's lower viscosity and longer hydrocarbon chains made it less adhesive to complex surfaces.

An equal weightage of 33.33% was given to each measured aspect to avoid bias toward any single measurement and provide a balanced evaluation of each coating's overall effectiveness in preventing corrosion. This suggests that while store-bought chemicals aimed at preventing corrosion are more effective, natural household alternatives like coconut oil can perform nearly as well and provide environmental benefits. Coconut oil is biodegradable and plant-derived, avoiding the sustainability issues associated with crude-oil-based products like WD-40 and chain lubricant, which contribute to greenhouse gas emissions during extraction.

Addressing the research question, WD-40 was the most effective coating at preventing corrosion of galvanized steel screws when exposed to 0.1 mol dm^{-3} sulfuric acid, based on percentage changes in length, width and mass.

A strength of this investigation is testing synthetic and household coatings under consistent conditions, allowing for a comparative evaluation of sustainable alternatives to commercial products. Measurements across three physical dimensions also provided a detailed assessment of corrosion.

Evaluation

Source of error	Type of error	Future improvements
Outliers	Random error	Conduct more trials (use more screws) to minimise the random error.
Potential unequal coating	Systematic error	Screws should be air dried for 2 hours rather than patted to ensure that the coating sticks and that all screws face equal conditions. Tissue fragments can be left on the screw and/or that more coating may be wiped out than other screws if patted, making the data less accurate and reliable.
Electronic balance fluctuations and accuracy	Random error	The electronic balance may give different readings due to air in the room. Next time a covered electronic balance which is accurate to more than two decimal places should be used to get more accurate data.
Evaporation of sulfuric acid	Systematic error	As all the beakers were kept in a fume cupboard, some of the sulfuric acid may have evaporated reducing its effective concentration and slow down corrosion. This could distort how effectively each coating prevents corrosion from 0.1 mol dm^{-3} sulfuric acid. In the future, all the beakers should be covered to ensure no sulfuric acid evaporates. The acidity of the solution can be measured, allowing for more detailed results as to which coating is most effective at preventing corrosion.
Lack of sulfuric acid concentration accuracy	Systematic error	Perform a titration to determine the concentration of the sulfuric acid to ensure accuracy, even if bottles were labelled incorrectly.
Did not measure mass of zinc sulfate produced	Systematic error	Weight the zinc sulfate produced from the reaction to then apply in the stoichiometric equation to predict and compare the mass of screw that was corroded.
Comparative experiment does not draw straightforward conclusion	Systematic error	Use different pH of sulfuric acid and temperatures the screws are exposed to. This ensures that the coating is universally effective rather than relatively better in the current setup.

Table 9. Limitations and weaknesses of the experiment, including the source of the error, the type of error and how to improve it for the future.

References

- Brown, C., Ford, M., Canning, O., Economou, A., & Irwin, G. (2023). *Higher Level Chemistry* (3rd ed., p. 312). Pearson Education. (Original work published 2023)
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Appendix

Screw Number	Coating	Initial Length	Final Length	Initial Width	Final Width	Initial Mass	Final Mass
1	Vegetable Oil	37.38	37.14	7.41	7.28	2.20	1.95
2	Vegetable Oil	37.35	37.06	7.31	7.19	2.19	1.94
3	Vegetable Oil	37.66	37.34	7.39	7.29	2.23	1.98
4	Vegetable Oil	37.39	37.12	7.34	7.22	2.22	1.97
5	Vegetable Oil	37.40	37.10	7.48	7.39	2.21	1.96
6	WD-40	37.34	37.02	7.28	7.23	2.21	1.95
7	WD-40	37.54	37.22	7.50	7.45	2.21	1.96
8	WD-40	37.59	37.27	7.47	7.42	2.25	1.99
9	WD-40	37.35	37.02	7.40	7.35	2.22	1.96
10	WD-40	37.37	37.01	7.37	7.32	2.19	1.93
11	Chain Lubricant	37.39	37.16	7.50	7.39	2.22	1.96
12	Chain Lubricant	37.27	37.04	7.41	7.29	2.19	1.93
13	Chain Lubricant	37.60	37.37	7.55	7.46	2.19	1.93
14	Chain Lubricant	37.30	37.03	7.37	7.26	2.21	1.96
15	Chain Lubricant	37.37	37.10	7.37	7.25	2.21	1.95
16	Coconut Oil	37.22	36.99	7.37	7.24	2.15	1.91
17	Coconut Oil	37.64	37.44	7.51	7.39	2.20	1.96
18	Coconut Oil	37.23	37.02	7.48	7.34	2.19	1.95
19	Coconut Oil	37.70	37.46	7.44	7.28	2.23	1.98
20	Coconut Oil	37.07	36.83	7.55	7.42	2.20	1.96
21	Control	37.43	37.12	7.56	7.38	2.17	1.93
22	Control	37.10	36.74	7.64	7.47	2.23	1.95
23	Control	37.30	36.98	7.48	7.28	2.18	1.92
24	Control	37.39	37.09	7.41	7.21	2.21	1.95
25	Control	38.88	38.51	7.81	7.63	2.20	1.94